

15	When a spring-mass system is displaced by an initial displacement X and released, it moves in a simple harmonic motion with a maximum momentum of p. The same spring-mass system is now displaced by an initial displacement of $0.5X$. What is the new maximum momentum of its motion, in terms of p?											
	A	$0.25p$	B	$0.5p$	C	$2p$	D	$4p$				
	Solution: B The maximum velocity is given by $v = \omega x_0$ where ω is the angular frequency and x_0 is the amplitude of the oscillation, which, in this case, is the initial displacement X. Also, since $p = mv$ Then, $p \propto v \propto X$ So, when the displacement is halved, the maximum momentum is halved as well.											

16	A transverse water wave is moving along the surface of some water. This causes a ball to
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